SOFTWARE ENGINEERING
& PROJECT MANAGEMENT

# DAY 11

# Requirements Elicitation Techniques & Stakeholder Analysis

Student Book • Module II: Requirements Engineering

**Learning focus**
Discover stakeholder needs systematically, select appropriate elicitation techniques, ask effective questions, and record findings without confusing assumptions with confirmed requirements.

**Running case study:** College Event Management System (CEMS)

Designed for classroom projection, discussion, practical application, and revision.

# 1. Learning Outcomes

By the end of this session, you should be able to:

• Explain why requirements elicitation is more than simply asking users what features they want.

• Identify stakeholders and analyse their influence, interest, needs, and possible conflicts.

• Choose suitable techniques such as interviews, questionnaires, observation, document analysis, workshops, and prototyping.

• Prepare open-ended, neutral, and follow-up questions.

• Record findings, assumptions, constraints, and unresolved questions for later analysis.

**Core idea:** elicitation is the disciplined discovery of needs and constraints from people, documents, existing systems, and the environment in which the software will operate.

# 2. What Is Requirements Elicitation?

Requirements elicitation is the activity of discovering, gathering, and clarifying what stakeholders need from a proposed system. The word *elicitation* is important: analysts do not only collect statements; they help stakeholders explain their goals, current work, difficulties, rules, exceptions, and expectations.

## Why it can be difficult

• Stakeholders may know the problem but not the best software solution.

• Different groups may use the same word to mean different things.

• People can forget unusual situations until they are asked about examples.

• Some needs are hidden in spreadsheets, paper forms, policies, or informal workarounds.

• Stakeholders may disagree about priorities, responsibilities, or acceptable risk.

• Needs can evolve as users see examples or learn what is feasible.

**Example:** A coordinator says, “We need an event registration page.” Follow-up questions may reveal that the real problems are duplicate registrations, unknown seat availability, and last-minute event changes. A page alone may not solve all three problems.

# 3. Identify the Stakeholders

A stakeholder is a person, group, or organisation that can affect the system, is affected by it, or has a legitimate interest in its outcome. Do not identify only the person who requested the project.

| Stakeholder | What to discover | Possible CEMS concerns |
|---|---|---|
| Students | Goals, access needs, devices, common difficulties | Find events, register quickly, receive accurate confirmation, understand cancellation rules |
| Faculty coordinators | Current workflow, event rules, reporting needs | Create events, set capacity, manage registrations, communicate changes |
| College administration | Policies, approvals, outcomes, accountability | Approve events, control roles, view reports, ensure policy compliance |
| IT support | Technology, operations, support, integrations | Account management, backups, availability, troubleshooting |
| Data/privacy owner | Data access, retention, consent, policy | Limit access to personal details and handle records appropriately |
| Accessibility representatives | Barriers and inclusive access | Ensure key tasks can be completed by users with different access needs |

## Stakeholder identification questions

• Who will use the system directly?

• Who approves the process, budget, policy, or release?

• Who maintains the system and handles incidents?

• Whose data is collected, stored, changed, or shared?

• Who is affected if the system is unavailable or produces incorrect results?

• Are there external services, vendors, or regulators whose rules matter?

# 4. Stakeholder Analysis

Once stakeholders are identified, analyse their needs and involvement. A stakeholder register helps the team remember who must be consulted, informed, or asked to approve decisions.

| Field | What to record | Example |
|---|---|---|
| Stakeholder / group | Name or role, not just an individual name | Faculty event coordinator |
| Goals and needs | What outcome matters to them | Prevent overbooking and track registrations |
| Influence | Ability to affect scope, decisions, or adoption | High: approves event workflow |
| Interest / impact | How strongly they are affected | High: uses the system every week |
| Knowledge / concerns | Rules, risks, assumptions, pain points | Cancellation deadline is not yet agreed |
| Engagement plan | How and when to involve them | Interview, then review prototype and SRS |

## Power–interest matrix

| Group | Suggested engagement | CEMS example |
|---|---|---|
| High influence, high interest | Work closely; involve in key decisions | Project sponsor, lead coordinator |
| High influence, lower day-to-day interest | Keep satisfied; provide concise decision updates | Senior administrator |
| Lower influence, high interest | Keep informed; gather frequent feedback | Students who will use registration |
| Lower influence, lower interest | Monitor and consult when relevant | Occasional event visitors |

> This matrix is a planning aid, not a way to dismiss people. A group with little formal authority may still experience major consequences from a design decision. Revisit the analysis when scope or stakeholder circumstances change.

# 5. Technique 1 — Interviews

An interview is a guided conversation with a stakeholder. It can be structured (same prepared questions), semi-structured (planned questions plus follow-ups), or relatively open-ended.

## When to use interviews

• To understand a workflow in depth.

• To explore motivations, exceptions, and concerns.

• When the stakeholder has specialist knowledge or decision authority.

## Example interview plan: faculty coordinator

• How do you announce an event today?

• Walk me through the registration process from start to finish.

• What information do you need before publishing an event?

• What happens when the event reaches capacity?

• Can registrations be cancelled or transferred? What rules apply?

• What reports do you prepare, and who uses them?

• Tell me about the last time the current process caused a problem.

## Interview good practices

• Explain the purpose and ask permission to take notes.

• Use open questions first, then specific follow-ups.

• Ask for real examples rather than accepting broad claims only.

• Summarise what you heard and ask the stakeholder to confirm.

• Separate what the stakeholder said from your own interpretation.

# 6. Technique 2 — Questionnaires and Surveys

Questionnaires collect responses from many people using a consistent set of questions. They are useful for identifying patterns across a large population, but they usually provide less depth than interviews.

## Example survey questions for students

• How do you currently find out about college events? (Select all that apply.)

• Which information is most important before registering: date, venue, capacity, eligibility, or organiser?

• Have you experienced duplicate or unsuccessful registrations? (Never / Sometimes / Often)

• Which notification method do you prefer? (Email / Portal notification / Other)

• Describe one improvement that would make event registration easier.

**Tip:** combine a few clear closed questions with a small number of open questions. Avoid double-barrelled questions such as “Is the site fast and easy to use?” because a person may have different answers for speed and ease of use.

# 7. Technique 3 — Observation

Observation means watching users perform their current work. It can reveal steps people forget to mention, unofficial workarounds, delays, repeated data entry, and environmental constraints.

## CEMS observation checklist

• How does a coordinator receive and approve an event request?

• Where is event information entered and stored?

• How are duplicates identified?

• How is remaining capacity calculated?

• What happens when details change after registration begins?

• Which steps require copying information between forms or spreadsheets?

Observe respectfully, obtain required permission, and protect confidential information. The goal is to understand the work, not to judge the person performing it.

# 8. Technique 4 — Document and Existing-System Analysis

Existing artefacts often contain business rules and data definitions. Review them alongside stakeholder conversations, because documents may be outdated or may not describe how work is actually performed.

| Artefact | What it may reveal |
|---|---|
| Registration form | Required fields, validation rules, consent, data collected |
| Spreadsheet or report | Data fields, calculations, reporting needs, duplicate handling |
| Policy document | Eligibility, approval, retention, access, and cancellation rules |
| Email or announcement template | Information students expect and communication timing |
| Existing application | Current workflows, integrations, limitations, and terminology |

Record the document name, owner, date/version, and any uncertainty. Confirm important rules with the responsible stakeholder rather than treating every old document as authoritative.

# 9. Technique 5 — Workshops

A workshop brings several stakeholders together to explore requirements, map a process, prioritise needs, or resolve differences. A facilitator helps everyone contribute and records decisions.

### A 30-minute workshop outline

| Time | Activity | Output |
|---|---|---|
| 0–5 min | Agree on the problem and goal | Shared objective |
| 5–12 min | Map the current process | Current-state workflow |
| 12–20 min | Identify pain points and desired outcomes | Problem list and candidate requirements |
| 20–26 min | Discuss conflicts and priorities | Decisions and open questions |
| 26–30 min | Read back actions and owners | Action list and follow-up plan |

Workshops are useful for cross-functional decisions, but they need facilitation. Senior voices should not dominate; invite quieter participants and record unresolved disagreements instead of forcing artificial agreement.

# 10. Technique 6 — Prototypes and Scenarios

A prototype is an early representation of a proposed system. It may be a paper sketch, wireframe, clickable mock-up, or limited working version. A scenario describes how a user and system interact to achieve a goal.

### CEMS prototype questions

• Can students identify event date, venue, eligibility, and available capacity?

• Is the registration action clear?

• What confirmation should appear after successful registration?

• How should a full event or failed request be presented?

• Can users find cancellation rules before they register?

A prototype helps stakeholders react to something concrete. However, a polished-looking screen can create the mistaken impression that the system is nearly complete. Label prototypes clearly and explain what is and is not implemented.

# 11. Choosing the Right Technique

| Need / situation | Useful technique | Reason |
|---|---|---|
| Understand a complex individual workflow | Interview + observation | Combines explanation with direct evidence |
| Collect preferences from hundreds of students | Questionnaire | Consistent responses at scale |
| Discover hidden steps and workarounds | Observation | Shows real behaviour in context |
| Find rules in existing forms and policies | Document analysis | Reuses available evidence and terminology |
| Resolve conflicting department expectations | Workshop | Enables joint discussion and decisions |
| Clarify a confusing user journey | Prototype + scenario walkthrough | Makes the interaction tangible |

Strong elicitation usually combines techniques. For example, survey students for broad patterns, interview a sample to understand why, observe the current process, and validate the findings with a workshop.

# 12. From Notes to Candidate Requirements

Elicitation findings are not automatically approved requirements. Classify each finding and track its status.

| Finding from elicitation | Interpretation / next step |
|---|---|
| “I often register twice because I do not know whether the first attempt worked.” | Candidate requirement: show a clear result and prevent duplicate confirmed registrations. Confirm expected duplicate-handling rules. |
| “We keep a spreadsheet to count available seats.” | Problem / current workaround: investigate capacity rules and identify a reliable source of truth. |
| “Students should get updates when a venue changes.” | Candidate requirement: define which changes trigger notification, who receives it, and through which channel. |
| “The website must be modern.” | Vague opinion: ask what problem this means—mobile layout, accessibility, navigation, visual style, or something else. |

### Useful elicitation record

• Finding ID and date

• Stakeholder or source

• Exact statement or observation

• Analyst interpretation

• Candidate requirement ID, if appropriate

• Assumption or unresolved question

• Owner and next action

## 13. Common Elicitation Mistakes

• Talking only to the sponsor and missing end users or support teams.

• Asking leading questions that suggest the answer.

• Jumping to screens or technology before understanding the problem.

• Recording assumptions as confirmed facts.

• Ignoring exceptions, error paths, accessibility, privacy, and operational needs.

• Failing to confirm notes and decisions with stakeholders.

• Using only one technique when different sources could reveal different aspects of the problem.

## 14. Classroom Activity — Plan an Elicitation Study

**Scenario:** students report missed event announcements and duplicate registrations. Coordinators say capacity tracking takes too much time.

• List at least six stakeholders or stakeholder groups.

• Choose three elicitation techniques and justify why each is suitable.

• Write six interview questions: at least four open-ended and two follow-up questions.

• Draft four survey questions for students.

• Identify two documents or existing artefacts to review.

• Create a stakeholder register with goals, influence, interest, and engagement approach.

• Write three candidate requirements and label any unresolved assumptions.

**Expected deliverable:** a two-page elicitation plan, stakeholder register, interview guide, sample survey, and initial findings log.

## 15. Knowledge Check

**1. What is requirements elicitation?**

The systematic discovery and clarification of stakeholder needs and constraints.

**2. Why should an analyst identify more than direct users?**

Approvers, maintainers, data owners, and affected groups may impose important needs or constraints.

**3. Which technique is especially useful for seeing workarounds?**

Observation, often combined with interviews.

**4. What is a weakness of a questionnaire?**

It usually provides less depth and context than an interview.

**5. What does a stakeholder register record?**

Stakeholders, their goals and concerns, influence/impact, and engagement approach.

**6. Why combine elicitation techniques?**

Different techniques reveal different evidence; combining them helps cross-check findings.

**7. What should happen when two stakeholders disagree?**

Record the conflict, clarify goals and constraints, facilitate analysis/negotiation, and document the decision or open issue.

**8. Is “the system must be modern” a ready-to-build requirement?**

No. It is ambiguous and needs clarification into specific, necessary, verifiable needs.

# 16. Glossary

| Term | Meaning |
|---|---|
| Elicitation | Discovering and clarifying stakeholder needs and constraints. |
| Stakeholder | A person or group affected by or able to influence the system. |
| Stakeholder register | A record of stakeholders, their needs, influence, concerns, and engagement plans. |
| Interview | A guided conversation used to explore requirements in depth. |
| Observation | Watching work as it happens to understand actual processes and constraints. |
| Workshop | A facilitated group session for discovery, analysis, prioritisation, or decisions. |
| Prototype | An early representation of a system or interaction used to gather feedback. |
| Assumption | Something treated as true for planning but not yet confirmed. |
| Open question | A question that invites explanation rather than a yes/no response. |

# 17. Homework

• Choose CEMS or another system approved by your instructor.

• Prepare a stakeholder register for at least six stakeholder groups.

• Create an interview guide with eight questions and a student survey with five questions.

• Select two additional elicitation techniques and explain when you would use them.

• Record five candidate requirements and identify the evidence/source for each.

• List three assumptions or unanswered questions that must be resolved before the SRS is finalised.

**Next session:** Requirement analysis, prioritisation, ambiguity, conflicts, and preparing requirements for the SRS.